|INCH-POUND|
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17 September 1989
SUPERSEDING
MIL-V-81995/7(AS)
17 MAY 1985

MILITARY SPECIFICATION SHEET

VALVE, STARTER CONTROL AIRCRAFT ENGINE, FOR T700 ENGINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the starter control valve described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specification and Standards (DODISS) specified in the solicitation: MIL-V-81995(latest revision).

REQUIREMENTS:

Envelope: See Figures 1 and 2

Ⓐ Weight: 3.9 lb maximum

Aircraft Model: SH-60

Engine Model: T700

Inlet Air Conditions

Rated Pressure: 45 psia

Rated Temperature: 450°F (232°C)

Ⓐ Maximum Pressure: 73 psig

Maximum Temperature: 480°F (250°C)

Ⓐ Maximum Engine Interbleed Pressure: 73psig

Minimum Pressure: 17 psig

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Regulating Pressure: 30±2 psig at inlet pressures to 51 psig at airflows from 15 lb/min to 30 lb/min. Opening pressure overshoot shall not exceed 2 psi.

Flow: The flow rate for all test conditions shall be limited by a fixed orifice or nozzle which will produce an airflow of 30±5 lb/min at orifice and nozzle inlet conditions of 45±0.5 psia and 400±25°F (204±14°C).

Rate of Pressure Rise: The initial pressure rise shall be limited to a maximum of 22 psi/sec during the first 0.5 second after the first indication of pressure rise and thereafter the pressure rise shall be smooth and at a positive rate.

Opening and Closing Times: Open 2.5 seconds maximum; close 0.5 second maximum.

Ⓐ Pressure drop: Pressure drop shall not exceed 0.5 psi at standard sea level rated condition.

Ⓐ Leakage: Internal shall not exceed 0.3 lb/min and total shall not exceed 0.5 lb/min when subjected to inlet air at 73 psig at 480±30°F (250±16°C).

Altitude:

a. Maximum operating: 20,000ft

b. Maximum non-operating: 50,000ft

Ambient Temperature Range:

a. Operating: -65 to +160°F (-54 to +71°C)

b. Non-operating: -65 to +320°F (-54 to +160°C)

Flight and Ground Loading:

a. Flight maneuvering: 20g

b. Shear: 50 lb

c. Axial: 150 lb

d. Moment: 300 in-lb

Ⓐ Vibration: The valve shall suffer no detrimental effects after being subjected to the vibration procedure specified in MIL-STD-810D, Method 514 category 4, test procedure I, and test conditions I.3.2.4.
Operation of the valve during vibration shall be required.The valve shall be pressurized to the maximum pressure inlet condition throughout the vibration test and any tendency for the valve to open without being activated shall be cause for rejection. In addition, a dwell at 17.2 Hz shall be conducted for 30 minutes in each of the three mutually perpendicular axis.

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Military Part Number: The military part number of the valve shall be M81995/7.

Interchangeability: The ATSCV shall be physically interchangeable or replaceable with any other manufacturer listed on the applicable Qualified Products List (QPL).

Useful Life: The valve shall have a minimum useful life of 8,000 aircraft hours or 8,000 engine start cycles whichever occurs first. The useful life is defined as that period of an item's operating life during which failures occur randomly and the failure rate is approximately constant.

Storage: The valve shall have a minimum total shelf life of five (5) years when stored as specified by the contractor under original packaging conditions. After such storage, the equipment shall be capable of meeting all requirements.

Position Indicator: The valve shall be equipped with an external position indicator which shall include suitable markings to indicate the valve "open" and "closed" positions. The valve shall also incorporate a microswitch in accordance with MIL-S-8805 which shall close when the valve butterfly is 6 degrees or more open. The switch shall have a 0.5 amp load rating at 28 vdc.

Ⓐ Order of precedence. In case of conflict between the text of this document and the reference documents cited herein, the text of this document shall take precedence.

Changes from previous issue. The margins of this specification are marked with letter Ⓐ to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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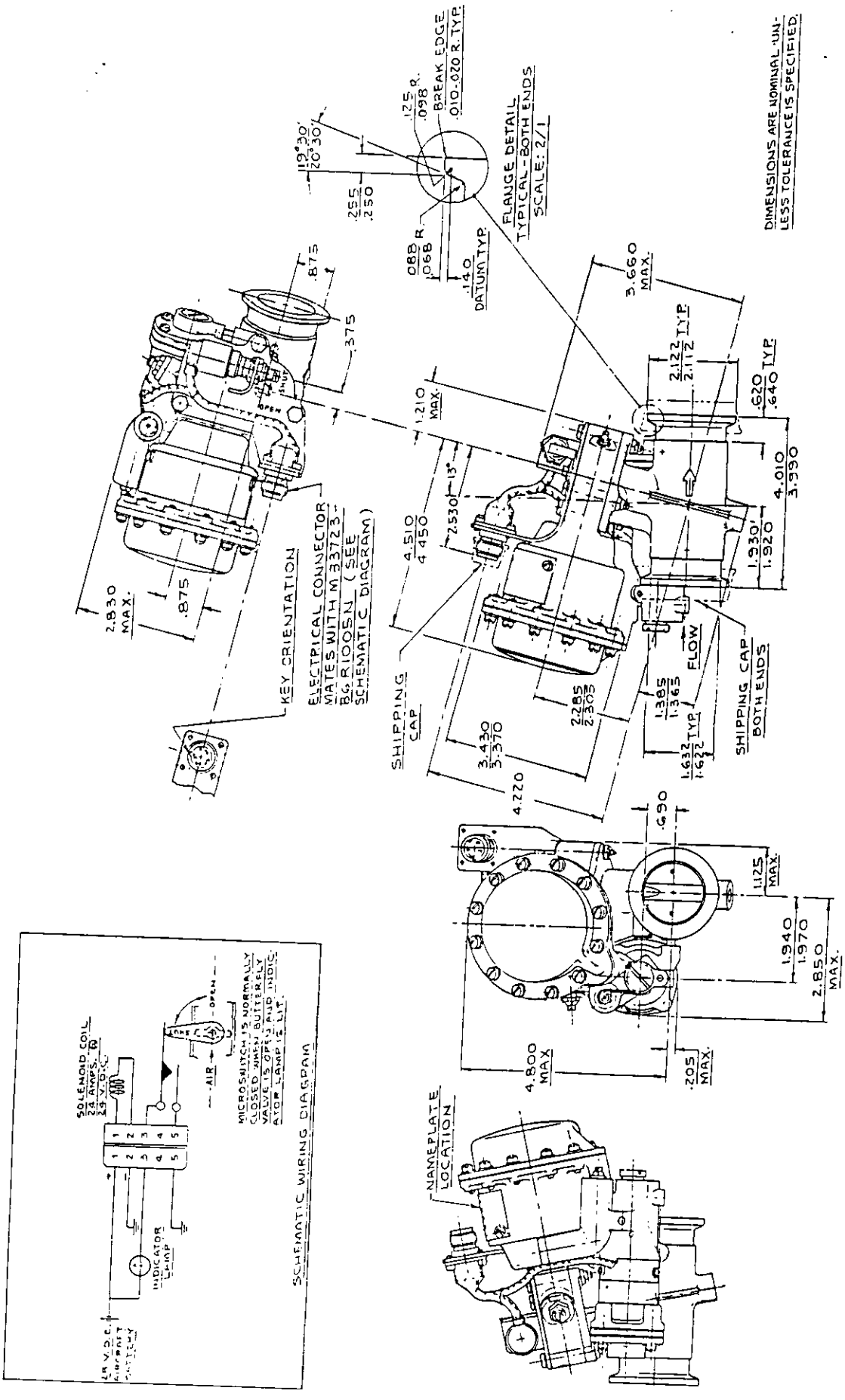


FIGURE 1. Starter control valve envelope - style 1.

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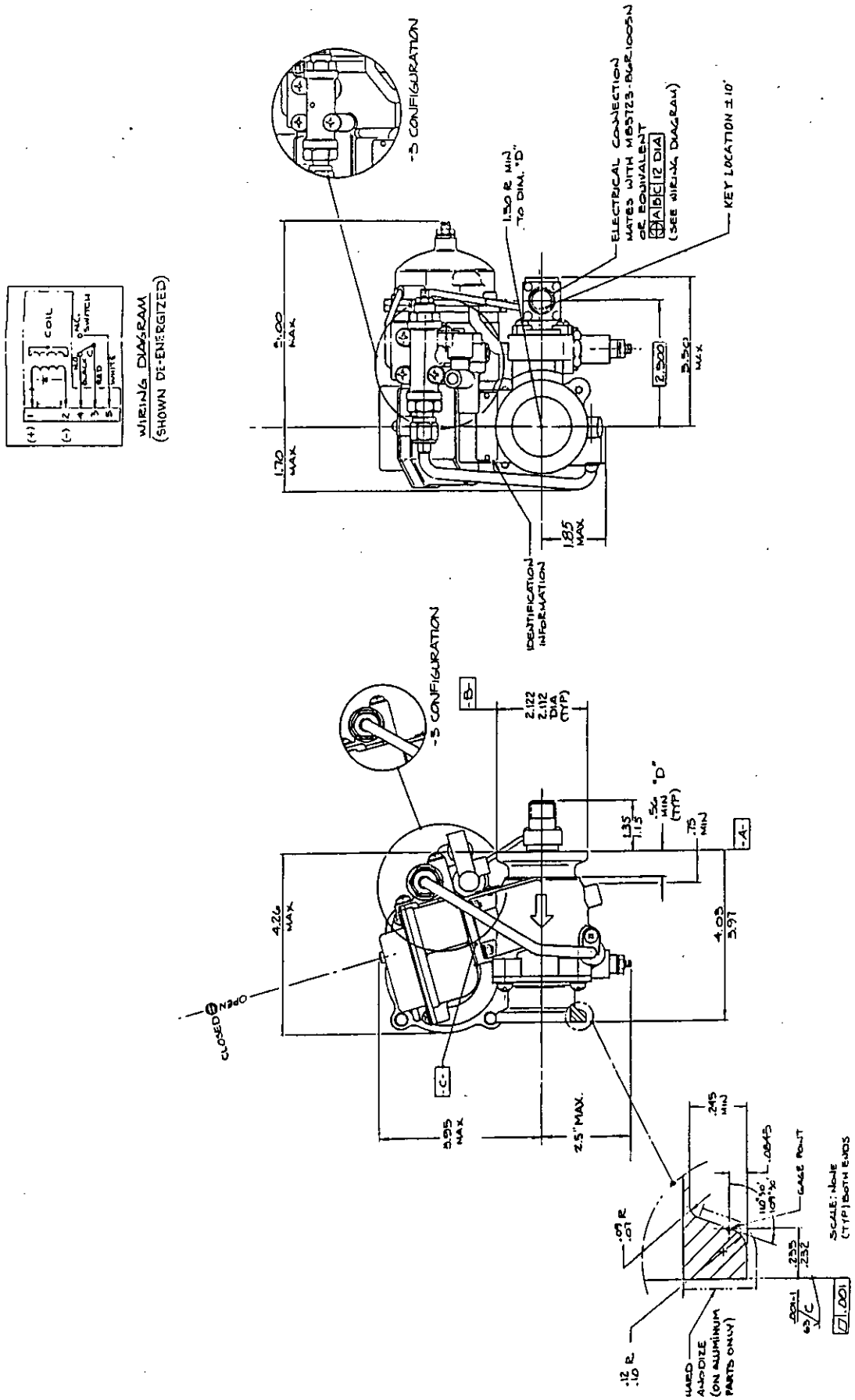


FIGURE 2. Starter control valve envelope - style 2.

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